Natural Language as a User-Facing Representation for Robotics

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Abstract

Robotic systems have traditionally relied on human-designed representations that make tasks and environments both explicit and computationally accessible. Recent advances suggest a different possible approach: natural language can itself serve as a shared, user-facing representation through which people specify and refine task-relevant knowledge. We argue that this role is particularly useful for concepts that are contextual, underspecified, or difficult to capture with fixed definitions or decision boundaries. We illustrate this idea through a disaster-response application in which a vision-language model must distinguish hazardous conditions from situations that are merely anomalous, a distinction whose operational meaning can be refined interactively through language. This process is a form of knowledge engineering: revising natural-language specifications changes how the system interprets the physical world. In this short paper, we suggest characterizing natural language as a representational layer in its own right, and describe language-based representations as a possible alternative to fully specified, human-crafted representations, particularly while task boundaries are being established. This perspective suggests that language can serve not only as an interface to robotic systems, but as part of the shared representational machinery through which both humans and agents construct and refine their understanding of the physical world.

Introduction

As large language models are extended with additional input modalities such as vision, the core of what counts as a 'representation' for these systems becomes blurred. When a person asks a physical robot to perform a task using natural language, should the discussion of its representational schema refer to natural language inputs, the embeddings over which it is operating, or some combination? Every representation a computer uses, whether manual or learned, requires translating human understanding into some other form before the machine can act on it, often at several levels of abstraction. These models make that step implicit. From the human interactor's perspective, language can now serve as the representation directly. While models still have (and indeed must have) internal representations, NL can constitute the operative, human-visible representation through which task-relevant world knowledge is specified and revised. We take the position that this should change what representational research aims at: in addition to designing better intermediate structures for robot reasoning, we should also study how language works as a human-visible representation schema.

The questions raised by this language-as-representation approach are made more complex by the addition of the real, physical world that the representation must capture so robots can operate in it. People think in a combination of language and symbols (Perlovsky 2007), implying both that language can cognitively support reasoning (Boroditsky 2011; Gentner 2016, *inter alia*)[1] and that an additional understanding of the physical world is necessary (Bisk et al. 2020). The existence of VLMs (vision-language models) both validates this claim and meets this need, allowing a reasoner to operate and end users to interact based on visual descriptions of the world (Radford et al. 2021). Large pretrained models have made it possible to use language as a direct representation of human intent (Sahetai* et al. 2026) but tying it to the physical world requires the incorporation of additional senses. A VLM can take NL representations of human meaning and produce a contextual understanding of how they tie into the world (Florea et al. 2026; Wang et al. 2025); in this use case, language has become synonymous with the representation.

Commonly, researchers turn natural language into robot behavior in a variety of ways, such as via language-conditioned policies like RT-2 (Zitkovich et al. 2023), parsing into formal logics such as LTL (linear temporal logic) (Liu et al. 2023b), and interactive concept grounding through dialogue or world analysis (Pavlick 2023). Such representations are operable, but leave one question open: do human and robot mean the same thing by the words? For example, a formula like '$\Box \neg$*hazard*' looks precise, but the human and the robot may not agree on what "hazard" denotes, especially since its meaning shifts in different contexts. LLM translation into formal representations is already error-prone and degrades outside its training domain (Xu, Feng, and Miao

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[1]This view is not universal (Fedorenko, Piantadosi, and Gibson 2024; Li and Gleitman 2002). An alternative perspective is that language is primarily communicative rather than supportive of symbolic thought. From this perspective, language is less demonstrably suitable for supporting underlying reasoning layers, but can remain an appropriate representation for interaction.

2024), and a formula can look correct while encoding the wrong intent (Cohen et al. 2024). This gap motivates keeping natural language as the representation for as long as possible before committing to an explicit interpretation.

The choice of NL as an operable representation format is driven in part by the ambiguity inherent in communication: many concepts are underspecified, difficult to define (but easy to recognize), and/or (especially) context-specific. We illustrate this with a vignette from a disaster-response system (Eskandari et al. 2026), in which a drone surveys a space before first responders enter. Language serves as the intermediary for two such concepts: whether something is *hazardous* to humans, and whether it is *anomalous* in its environment (Indukuri et al. 2026).

> ***Vignette.*** As an example of both the user-facing capabilities provided by representing the physical world using NL, as well as some of the points of difficulty, we implement the following disaster response use case (each step is performed either by the **system** or by the **situation commander**):
>
> **Situation commander** wants to be made aware of hazards in the environment before first responders physically enter a potentially dangerous space. First, a drone is sent in to take videos of the environment.
>
> **System** must (1) identify and (2) convey points of interest given the commander's goals. With naïve prompts such as "What in the environment might be hazardous," two categories of things are returned: True *hazards* and non-hazardous but atypical *anomalies* (see Figure 2).
>
> **System** takes in human-understandable prompts to identify and differentiate these cases. It assesses individual frames of the video against the queries (see Prompt, below).
>
> **System** identifies both hazards (which can be dangerous to humans) and anomalies (which, if not also hazardous, may be distracting but are not immediately dangerous). A digital twin of the environment is constructed.
>
> **Situation commander** and/or first responders can, via a VR headset, explore the environment in virtual reality without risk (Figure 1, left). Hazards and anomalies are annotated with user-interactable popup descriptions.
>
> **Situation commander** can iterate on NL prompts and VLM responses to ask specific followup questions or improve the queries presented until satisfied with the outcome. For example, they may consider the specific hazards shown and specify "We are only interested in immediate hazards such as the chemical spill, not potential future hazards."
>
> **System** can add this statement to the context and adjust the VR environment to include only more immediately hazardous elements (Figure 1, right).

In this vignette, the representation contains concepts without crisp definitions; the human can directly inspect and modify those concepts, and modifying the linguistic representation changes the system's interpretation of the physical environment. These are representational operations, not interface operations. We clarify the representational layer in the section Language as a Representation. Because the representation is language, the commander can sharpen the queries until that line is drawn correctly, then explore the annotated results as a digital twin in VR. That loop works because the representation being refined is language itself.



Figure 1: *Left:* an example of a potential hazard annotated in the VR environment as worth human attention. *Right:* a hazard view after the situation commander iterated on their specific requirements.

Language as a Representation

It is useful to distinguish three layers of representation. At the outermost, the human–machine interface, natural language is an obvious choice: users speak or type requests to robots. At the innermost, which users see as a black box, models operate over learned embeddings. Our claim concerns the layer between them: the task-level representation over which the system reasons about the world, such as what counts as a hazard and which distinctions matter. We argue that this layer, which has been traditionally formal, can itself remain in natural language. Specifically, NL can encode task goals, distinguish related concepts (e.g., hazard vs. anomaly), supply background knowledge, and provide rules for interpreting sensor evidence. Of course, a robot may still require scene graphs, symbolic state descriptions, or other structured representations. We distinguish these machine-internal representations from a *shared* representation: the form in which the system and human express, inspect, and revise task-relevant knowledge. Our claim is that with recent advances in language models, natural language is becoming an increasingly viable candidate for this role.

Language need not simply be an input that is translated into another representation and then discarded; instead, it can persist, allowing people to specify intent, characterize the world, introduce conceptual distinctions, and revise those distinctions in response to system behavior. In our vignette, for example, the distinction between hazardous and merely anomalous is not initially captured by a fixed taxonomy or decision boundary. The user instead refines what those concepts mean for the task through language, and the system applies that evolving specification to perceptual observations of the physical environment. Refining the language amounts to refining the task-relevant representation of that environment.

Natural language is particularly useful in this role when the concepts being represented are underspecified or not characterized by crisp membership criteria. Its ambiguity is consequently both a limitation and an affordance. Formal representations are appropriate where exact constraints, verification, or unambiguous execution are required, but requiring formalization too early can force distinctions that users have not yet made and discard context needed to make them. Rather than claiming that formal representations are unnecessary, we instead argue that language should be treated as a representational layer in its own right, particularly while

task concepts and their boundaries are being negotiated and grounded in the physical world.

Prompts as Knowledge Engineering

Framing natural language interactions as a desirable representation also supports the claim that prompt development is a form of knowledge engineering (Koutsiana et al. 2024), conducted largely in natural language with the end goal of inducing a particular knowledge state in the system. If prompts specify concepts, relationships, task context, and desired distinctions, and users iteratively revise those specifications in response to system behavior, then prompt refinement is effectively interactive construction of a task/world representation. This is consistent with the level of technical depth, learning, and effort that goes into designing VLM queries: although natural language reduces the need to learn a formally specified language, prompt design is still a skill that benefits from learned approaches (Liu et al. 2026). For an example of prompt engineering as knowledge engineering, the definitions above correspond not only to a particular set of meanings for terms, but also to specifying a problem the system must operate on (e.g., identifying points of interest to first responders, see Vignette).

> **Prompt.** You are an expert safety inspector specializing in hazard identification. Your primary role is to accurately and thoroughly identify potentially hazardous items in various environments. Provide clear, detailed explanations for why each item might be hazardous and suggest appropriate safety measures, ensuring high standards of safety are maintained. Respond concisely. *(formatting instructions follow)*

VLMs can achieve surprisingly strong decision-making performance when visual scenes are replaced by textual descriptions obtained via image captioning (Hu, Li, and Yin 2025; Driess et al. 2023), suggesting that foundational reasoning can be effectively learned from language. We hold that when a concept has no single obvious definition, the representation carrying that judgment should stay in natural language up to the point of human decision, rather than being collapsed into a class label or score early. A model's language output may already carry the distinction; reducing it to a formal signal risks discarding the context that made the judgment possible. This reframes the open question from "can a model tell hazard from anomaly" to "what representation lets that distinction survive deployment." We take the latter to be the more productive question, and the one our vignette is built to probe.

Ambiguity and Family Resemblances

In high-risk environments, VLM queries often define the concept of anomaly vaguely, which can lead to confusing harmless anomalies with hazards (Indukuri et al. 2026) (see Figure 2 for examples). Wittgenstein observed that some words have no simple membership criteria; for example, there is no trait that all games have and only games have: board games, ball games, and solitaire only share overlapping similarities that create a *'family resemblance'*, yet we use the word without difficulty (Wittgenstein 1953). The term



Figure 2: Images showing actual hazardous situations *(left)* as opposed to situations that are merely anomalous *(right)* (Indukuri et al. 2026). VLMs have a tendency to confuse the two categories, returning all four images as hazards.

'hazardous' has similar characteristics (Cameron et al. 2017). A wet floor, exposed wiring, and a loose ceiling beam are dangerous for unrelated reasons. Given this behavior, there is no single property a model can check for when evaluating hazards from naïvely constructed descriptions, so it uses the consistent pattern from its training data that problematic cases look different from normal ones. In effect, "Is this hazardous?" gets answered as "Does this look unusual?" This may be the source of the hazard/anomaly collapse observed.

The more formal the language gets, the less clear it becomes for concepts like hazard, where humans themselves disagree on the wording. The difficulty then lies less in how carefully a hazard taxonomy is specified than in the fact that the boundary itself shifts. Formal representations are good at capturing well defined boundaries; the open question is how these boundaries get settled (for example, for a particular disaster site/responder pair, what hazards are interesting and worth surfacing?). Natural language is the most flexible medium for iterating on desired behavior, with the potential for formal structure to take over once the distinction has been drawn (for example, requiring the model to output LTL interpretations (Liu et al. 2023b); while this provides for the possibility of formal correctness checking, it is not always a requirement that examinable formal representations be used). We claim that using language to force VLMs to conceptually separate anomalous scenes from truly hazardous ones helps focus users' attention to more critical situations. To improve the ability of state-of-the-art VLMs in distinguishing hazards, anomalies, and benign scenes, it is possible to distinguish types of irregularities, corresponding to the following definitions (which have necessarily context-dependent interpretations):

- **Safe**: No immediate hazard is present and object-context relations are not violated.
- **Anomalous**: A scene element appears out of place or in-

consistent with expected relations (interposition, support, probability, position, or familiar size), with no immediate physical risk.

- **Hazardous**: A visible source of immediate harm is present, consistent with OSHA's definition of a hazard (Occupational Safety and Health Administration 2002), without any contextual irregularity.
- **Anomalous/Hazardous**: The scene presents both an immediate hazard and an object–context violation.

These definitions attempt to minimize ambiguity by calling out the characteristics that complicated, ambiguous concepts have in common. This approach is helpful to obtaining well-formed human annotations of images, but not necessarily sufficient guidance for a VLM. Instead, defining terms provides a starting point for NL-based, iterative improvements to the representation.

Limitations of Language as a Representation

A significant limitation of using language as a representation is that it is not formally examinable the way code or a logical formula is. There is no way to verify that the system has understood a prompt correctly or will behave in a demonstrably safe or correct way. While in some cases it is feasible to translate NL into something checkable using existing tools (Code as Policies (Liang et al. 2023), Lang2LTL (Liu et al. 2023b), LLM+P (Liu et al. 2023a), *inter alia*), formalizing too early may force a decision before it is actually appropriate. In our vignette, the commander is still working out what "hazard" should means via interaction; locking that into a formal representation too soon would remove the flexibility that made the iterative refinement useful. Nonetheless, this positive use of ambiguity must be balanced with the risks of having an underspecified or non-obviously incorrect interpretation of human-facing interactions.

In practice, current language-based systems are often built on LLMs and VLMs, with the usual associated failure modes: hallucination, "bullshitting" (Frankfurt 2009), and excessive sycophancy (Cheng et al. 2025)—any of which could lead to a breakdown in correctness, which could in turn lead to misalignment or even, as in our exemplar, danger. Since we propose that language itself serve as the representation, these failures are no longer confined to surface-level output; they can degrade the system's actual understanding of the situation, propagating directly into downstream decisions. These failure modes are in addition to broader ethical questions about the use of large pre-trained models (Jiao et al. 2025; Deng et al. 2025; Laakso, Kemell, and Nurminen 2024).

Future Directions

This view of natural language as a representational scheme suggests several directions for future effort and analysis:

1. It is useful to think of prompts—as a form of engineered knowledge—as being *evolving* and *contextual*, rather than being fixed at development time. Adapting to the specific application (e.g., different disaster/responder dyads) allows context-sensitive, iterative prompt refinement to be responsive to a particular scenario.
2. If we consider iterative development to be a desirable behavior, the model's responses are never final: instead, the user determines when results are sufficient. This permits both customization of model behavior and ongoing adaptation in dynamic environments.
3. If the query and represented background knowledge are not assumed to be fixed, there is value to allowing ongoing revision of a model's conclusions, both via human intervention and by using the models' own judgment about its conclusions, similar to self-refinement methods like Reflexion (Shinn et al. 2023).
4. We suggest moving away from implementation details like scalar thresholds (e.g., "Notify the user if the probability of something being hazardous exceeds 0.8") or pre-defined formatting requirements. Giving the model more unconstrained language to process early helps support human-robot interactions. The trade-off is auditability: a fixed threshold's trigger is known in advance and checkable after the fact, while a linguistic judgment is not. Where audit trails or compliance guarantees are required, an explicit, checkable threshold may need to accompany the language-based reasoning we describe.
5. In applications where it is difficult to come up with unambiguous definitions, reasoning may benefit from human-style 'family resemblance' models, in which the human interlocutor supplies concrete examples of concepts rather than definitions alone.

Taken together, these directions suggest a shift in how we think about representation for robots operating with people. Natural language is not solely an input that is translated into more formal internal representations; it can itself be a useful representational medium, particularly while concepts are still being negotiated. Its ambiguity, often treated primarily as a weakness, also provides flexibility: users can supply examples, introduce context, and refine earlier judgments using the same representational vocabulary that they use to express their intent. Formal representations remain valuable where precision and well-defined boundaries are required, but language provides the human-visible layer through which those boundaries are established. Treating that layer as a representation in its own right opens a rich set of questions about how human and machine reasoning can jointly construct an understanding of the physical world.

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